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MATHEMATICS

1.

2	210
5	105
3	21
7	7

5	55
11	11
	1

HCF = 5

$\Rightarrow (210 \times 5) + 55(y) = 5$

$\Rightarrow 55(y) = 5 - 1050$

$\Rightarrow y = -\frac{1045}{55}$

$\therefore y = -19$ (1)

2. We can conclude a and b are odd numbers
Now, let a be $(2x+1)$ for x being +ve integer
Similarly
Let b be $(2y+1)$ for y being +ve integer

$\Rightarrow a+b = 2x+1+2y+1$

$= 2x+2y+2$

$\therefore a+b = 2(x+y+1)$

So, $(a+b)$ is divisible by 2 (1)

3. (3) 20
4. (3) 476

5. $p(1) = 1-3+2+6$
$= 6$

$q(2) = \cancel{32} \cancel{-8} \cancel{+8} + 2 - 6$
$= 28$

$p(1) + q(2) = 6 + 28 = 34$ (1)

6. $D = 0 \Rightarrow b^2 - 4ac = 0$

$\Rightarrow (3k)^2 - 4(2)(8) = 0$

$\Rightarrow 9k^2 - 64 = 0$

$\Rightarrow k^2 = \frac{64}{9} \quad \therefore k = \pm\frac{8}{3}$ (4)

7. (1) $b - a + 1$
8. (1) 5
9. $D = -68 \Rightarrow b^2 - 4ac = -68$
Quad: $mx^2 - 4x + 7 \Rightarrow (-4)^2 - 4(m)(7) = -68$

$\Rightarrow 16 - 28m = -68$

$\Rightarrow 28m = 84$

$\therefore m = 3$ (4)

10. $x^2 - 2x - 3 = 0$ $\therefore \alpha+\beta = \frac{-b}{a} = 2$

$\alpha\beta = \frac{c}{a} = -3$

Now, Let $p = \alpha+2$ & $q = \beta+2$

Now, $p+q = \alpha+2+\beta+2$

$= \alpha+\beta+4$

$= 2+4$

$\boxed{p+q = 6}$

$pq = (\alpha+2)(\beta+2)$

$= \alpha\beta + 2\alpha + 2\alpha\beta + 4$

$= \alpha\beta + 2(\alpha\beta) + 4$

$= -3 + 4 + 4$

$pq = 5$

$\therefore$ req quad eq $= (x^2 - 6x + 5)k$ (2)

11. Let the two consequtive +ve integers be x & $x+1$

Then,

$x^2 + (x+1)^2 = 365$

$\Rightarrow x^2 + x^2 + 2x + 1 = 365$

$\Rightarrow 2x^2 + 2x - 364 = 0$

$\Rightarrow x^2 + x - 182 = 0$

Using Siddhatacharya's formula, we get

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1+728}}{2} = \frac{-1 \pm 27}{2}$$

either $x = 13$ or $x = -14$
(not possible)

$\therefore$ The two consequtive +ve integers are 13 and 14

12. $p(x) = x^2 - p(x+1) - k$

$\Rightarrow p(x) = x^2 - px - p - k$

Since α and β are zeroes

$\therefore \alpha + \beta = \frac{-(-p)}{1} = p$

$\alpha\beta = \frac{-p-k}{1} = -p-k$

Given, $(\alpha+1)(\beta+1) = 6$

$\Rightarrow \alpha\beta + \alpha + \beta + 1 = 6$

$\Rightarrow \cancel{p} - \cancel{p} - k + 1 = 6$

$\therefore \boxed{k = -5}$

13. Let Rehman's present age be x yrs

3 yrs ago, Rehman's age $= (x-3)$ yrs

$\Rightarrow$ Reciprocal $= \frac{1}{x-3}$ —(i)

5 yrs hence, his age $= x+5)$ yrs

$\Rightarrow$ Reciprocal $= \frac{1}{x+5}$ —(ii)

A/Q, $\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3}$

$\Rightarrow \frac{(x+5+x-3)}{(x+5)(x-3)} = \frac{1}{3}$

$\Rightarrow \frac{2x+2}{x^2+2x-15} = \frac{1}{3}$

$\Rightarrow x^2 + 2x - 15 = 6x + 6$

$\Rightarrow x^2 - 4x - 21 = 0$

$\Rightarrow x^2 - 7x + 3x - 21 = 0$

$\Rightarrow x(x-7) + 3(x-7) = 0$

$\Rightarrow (x-7)(x+3) = 0$

either $x = -3$ (not possible) | or $x = 7$

$\therefore$ Present age of Rehman $= 7$ yrs

14. Required minimum distance = LCM(40, 42, 45)

5	40, 42, 45
2	8, 42, 9
2	4, 21, 9
2	2, 21, 9
3	1, 21, 9
3	7, 3
7	7, 1
	1, 1

LCM (40, 42, 45) = 2520 cm

15. Let $\sqrt{5}$ be rational number

(i) i.e. $\sqrt{5} = \frac{p}{q}$ where p and q are coprime & $q \neq 0$

$\Rightarrow \sqrt{5}q = p$

(Sq. both sides)

$\Rightarrow 5q^2 = p^2$ —(i)

So, p^2 divisible by 5

p is divisible by 5

Let $p = 5c$

(Sq. both sides)

$p^2 = 25c^2$ —(ii)

putting p^2 in eq-(i)

$5q^2 = 25(c)^2$

$\Rightarrow q^2 = 5c^2$

So, q is divisible by 5

Thus p and q have common factor 5 other than 1 & itself.

Thus our assumption that p & q are co prime is wrong.

This contradicts our assumption.

$\therefore \sqrt{5}$ is an irrational number.

15. (ii) Let us consider a +ve integer x.
Divide the +ve integer x by 3 and let r be the remainder and q be the quotient such that –

$x = bq + r$ [Acc. to Euclid's division lemma]
$x = 3q + r$ — (i) $(0 \le r < b)$

So, r is an integer which lies b/w 0 and 3
Hence, r can either be 0, 1 or 2

Case – I

when $r = 0$ – $x = 3q$
$\Rightarrow x^2 = (3q)^2$ [Sq. both sides]
$\Rightarrow x^2 = 3(3q^2)$
$\Rightarrow x^2 = 3m$ where $3q^2 = m$

Case – II

when $r = 1$ – $x = 3q + 1$
$\Rightarrow x^2 = (3q+1)^2$ [Sq. both sides]
$\Rightarrow x^2 = (3q)^2 + (1)^2 + 2(3q)(1)$
$\Rightarrow x^2 = 9q^2 + 6q + 1$
$\Rightarrow a^2 = 3(3q^2 + 2q) + 1$
$\Rightarrow a^2 = 3m + 1$
where $(3q^2 + 2q) = m$

Case – III

where $r = 2$ – $x = 3q + 2$
$\Rightarrow x^2 = (3q)^2 + 2^2 + 2(3q(2)$
[Sq. both sides]
$\Rightarrow x^2 = 9q^2 + 12q + 4$
$\Rightarrow x^2 = 3(3q^2 + 4q + 1) + 1$
$\Rightarrow x^2 = 3m + 1$
where $(3q^2 + 2q + 1) = m$

(Hence proved)

16. (i)

$$\begin{array}{r|l} & \quad 2x^2-2x-1 \\ 2x^2+x-2 & 4x^4-2x^3-6x^2+x-5 \\ & -\;\underset{(-)}{4x^4}\;\underset{(+)}{+2x^3}\;-4x^2 \\ \hline & \quad -4x^3-2x^2+x \\ & -\;\underset{(+)}{-4x^3}\;\underset{(+)}{-2x^2}\;\underset{(-)}{+4x} \\ \hline & \quad -2x^2-3x-5 \\ & -\;\underset{(+)}{-2x^2}\;\underset{(+)}{-x}\;\underset{(-)}{+2} \\ \hline & \quad -2x-7 \end{array}$$

Hence, $(-2x-7)$ should be subtracted from $4x^4-2x^3-6x^2+x-5$ so that result is exactly divisible by $2x^2+x-2$

(ii) zeroes of polynomial are $2 \pm \sqrt{3}$

Thus factors are $(x-2+\sqrt{3})$ and $(x-2-\sqrt{3})$

Now,

$(x-2+\sqrt{3})(x-2-\sqrt{3}) = x^2+4-4x-3$

$= x^2-4x+1$

Thus, x^2-4x+1 is a factor of the given polynomial

Now,

$$\begin{array}{r|l} & \quad x^2-2x-35 \\ x^2-4x+1 & x^4-6x^3-26x^2+138x-35 \\ & -\;x^4\;\underset{(+)}{-4x^3}\;\underset{(-)}{+x^2} \\ \hline & \quad -2x^3-27x^2+138x-35 \\ & -\;\underset{(+)}{-2x^3}\;\underset{(-)}{+8x^2}\;\underset{(+)}{-2x} \\ \hline & \quad -35x^2+140x-35 \\ & -\;\underset{(+)}{-35x^2}\;\underset{(-)}{+140x}\;\underset{(+)}{-35} \\ \hline & \quad 0 \end{array}$$

$\therefore$ $x^4-6x^3-26x^2+138x-35 = (x^2-4x+1)(x^2-2x-35)$

Now,

$x^2-2x-35 = x^2-7x+5x-35$

$= (x-7)(x+5)$

$\therefore$ The value of the polynomial is also zero when $x-7=0$ or $x+5=0$

Hence 7 and (-5) are also zeroes of polynomial

17. (i) Max^m no. of participants in each room
= HCF (60, 84, 108)

$$\begin{array}{r|l} 2 & 60 \\ \hline 2 & 30 \\ \hline 3 & 15 \\ \hline & 5 \end{array} \quad \begin{array}{r|l} 2 & 84 \\ \hline 2 & 42 \\ \hline 3 & 21 \\ \hline 7 & 7 \\ \hline & 1 \end{array} \quad \begin{array}{r|l} 2 & 108 \\ \hline 2 & 54 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$HCF = 2 \times 2 \times 3$
$= 12$ (2)

(ii) The minimum number of rooms required
$= \dfrac{60+84+108}{12} = \dfrac{252}{12} = 21$ (1)

(iii)
$$\begin{array}{r|l} 2 & 60, 84, 108 \\ \hline 2 & 30, 42, 54 \\ \hline 3 & 15, 21, 27 \\ \hline 3 & 5, 7, 9 \\ \hline 3 & 5, 7, 3 \\ \hline 5 & \cancel{5, 7, 1} \\ 7 & \end{array}$$

LCM = 3780 (2)

(iv) HCF × LCM = 12 × 3780 ~~45360~~
= 45360 (1)

(v)
$$\begin{array}{r|l} 2 & 108 \\ \hline 2 & 54 \\ \hline 3 & 27 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$$

$108 = 2^2 \times 3^3$ (1)

18. (i) (1) −2, 8

(ii) $x^2 - (\alpha+\beta)x + \alpha\beta$
$= x^2 - (-2+8)x + (-2)(8)$
$= x^2 - 6x - 16$
$= -x^2 + 6x + 16$ (2)

(iii) $-(4)^2 + 6(4) + 16$
$= -16 + 24 + 16 = 24$ (3)

(iv) $-x^2 + 3x - 2 = 0$
$\Rightarrow x^2 - 3x + 2 = 0$
$\Rightarrow x^2 - x - 2x + 2 = 0$
$\Rightarrow x(x-1) - 2(x-1) = 0$
$\Rightarrow (x-2)(x-1) = 0$ either $x = 2, x = 1$ (1)

(v) $\alpha+\beta = -3$ $\quad$ $\alpha\beta = -28$

$\Rightarrow 4+\beta = -3$

$\Rightarrow \beta = -7$

Now, req. polynomial $= k[x^2 - (\alpha+\beta)x + \alpha\beta]$

$= k[x^2 - (-3)x - 28]$

$= k[x^2 + 3x - 28]$

$= -x^2 - 3x + 28$

(2)